

Joaquim Cera Da Conceição

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Born on 06/06/1998

Education and affiliations

- 2025–2026 **ATER (Temporary Teaching and Research Associate) in mathematics**, Université de Caen Normandie (LMNO)
- 2022–2025 **PhD in Mathematics**, Université de Caen Normandie (LMNO),
Advisor: Christian Ballot
Thesis: Various density questions in the theory of Lucas sequences
- 2021–2022 **Master's Degree in Pure Mathematics**, *Number theory and functional analysis*, Université de Bourgogne Franche-Comté
- 2019–2021 **Master's Degree in General Mathematics**, Université de Caen Normandie
- 2016–2019 **Bachelor's Degree in Mathematics**, Université de Caen Normandie, 3-year program

Experience

- Mar.–Apr. 2025 **Visiting Scientist**, Max Planck Institute for Mathematics,
Mentor: Pieter Moree
- 2022 **Research Internship**, *Master's thesis*, Université de Caen Normandie,
Advisor: Christian Ballot
Title: Two density studies
- 2021 **Research Project**, Université de Franche-Comté,
Advisor: Christophe Delaunay
Title: The prime number theorem
- 2020 **Research Project**, Université de Caen Normandie,
Advisor: Christian Ballot
Title: Euler's totient function for Gaussian and Eisenstein integers
- 2019 **Laboratory Internship**, Université de Caen Normandie,
Advisor: Christian Ballot
Title: Solvable, nilpotent, abelian and cyclic numbers

Organisation

- 2025–ongoing **PhD Speed Talks**, *LMNO (Caen)*

Preprints

- 2026 **Explicit Density Computations for Certain Sets of Primes in Lucas Sequences**, Available at <https://arxiv.org/abs/2604.20014>

Publications

- 2025 **Divisibility of the Multiplicative Order Modulo Monic Irreducible Polynomials Over Finite Fields**, *Journal of Number Theory*, 277 (2025), 105–123
- 2025 **Primitive Divisors of Lucas Sequences in Polynomial Rings**, *Journal of Algebra*, 672 (2025), 400–412
- 2023 **Multinomial Catalan Numbers and Lucas analogues**, *Journal of Integer Sequences*, 26 (2023), Article 23.9.7

Teaching

- 2025–2026 **Calculus**, *first-year computer-science students*, equations and inequalities, functions, sequences, limits, asymptotic analysis
50h of mixed courses and exercise sessions
- Probability**, *first-year mathematics students*, sets and functions, sums and products, combinatorics, introduction to probabilities
50h of mixed courses and exercise sessions
- Applied linear algebra**, *third-year applied-mathematics students*, linear programming, matrix decomposition, singular value decomposition, simplex method
10h of practical sessions
- Linear Algebra**, *second-year mathematics students*, geometry, endomorphisms, Euclidean spaces, symmetry, projection
35h of exercise sessions
- Linear Algebra**, *first-year computer-science students*, systems of linear equations, real-coefficient polynomials, vector spaces
50h of mixed courses and exercise sessions
- 2024–2025 **Calculus**, *first-year computer-science students*, equations and inequalities, functions, sequences, limits, asymptotic analysis
50h of mixed courses and exercise sessions
- 2023–2024 **Approfondissement en Mathématiques**, *first-year mathematics students*, induction, logic, sets, complex numbers, differential equations
30h of exercise sessions
- Mock Oral Exams (“Colles”) in Approfondissement en Mathématiques**, *first-year mathematics CUPGE students*, 10h
- Linear Algebra**, *second-year mathematics students*, geometry, endomorphisms, euclidean spaces, symmetry, projection
15h of exercise sessions

Talks

- Jan. 12, 2026 **Sur le théorème de Loher**, Working Group on Lehmer’s conjecture (Caen, France)
- Sep. 18, 2025 **On the rank of appearance of polynomial Lucas sequences**, 9th Number Theory Meeting (Turin, Italy)

- June 30, 2025 **On the rank of appearance of polynomial Lucas sequences**, 33èmes Journées arithmétiques (Esch-sur-Alzette, Luxembourg)
- May 25, 2025 **Une conjecture d’Artin et des suites de (non-)torsion**, PhD students retreat, LMNO (Bayeux, France)
- May 22, 2025 **Sur le rang d’apparition des suites de Lucas polynomiales**, “Arithmétique” seminar, LPP (Lille, France)
- May 16, 2025 **Sur le rang d’apparition des suites de Lucas polynomiales**, Number Theory seminar, IMB (Bordeaux, France)
- Apr. 28, 2025 **Density questions on primitive divisors of Lucas sequences**, PLeaSANT seminar, MPIM (Bonn, Germany)
- Oct. 7, 2024 **Nombres de Catalan multinomiaux et suites de Lucas**, Young Researchers seminar, LMNO (Caen, France)
- July 11, 2024 **Divisibility of the rank of appearance of primes in Fibonacci polynomials in $\mathbb{F}_q[T]$** , 21st International Fibonacci Conference (Claremont, CA, USA)
- Nov. 23, 2023 **Prime divisors of companion Lucas sequences in $\mathbb{F}_q[\theta]$** , Number Theory team retreat, LMNO (Bayeux, France)
- May 17, 2023 **Quid of primes for which the multiplicative order of rationals is divisible by a fixed integer**, QUID seminar, IMT (Toulouse, France)
- Dec. 8, 2022 **Premiers pour lesquels le ppcm des ordres de nombres rationnels est multiple d’un entier fixé**, Young Researchers seminar, LMNO (Caen, France)

Training

- 2025-ongoing **Working Group on Lehmer’s conjecture**, *Around Lehmer’s conjecture: from zero characteristic to positive characteristic* (Caen, France)
 - 2025 **Research Ethics**, *Université de Lyon, France*
 - 2024 **Scientific Integrity**, *ED MIIS (Caen, France)*
- 2023–2024 **Working Group on Elliptic Curves**, *Young Researchers Working Group* (Caen, France)
 - 2023 **Summer School**, *local Langlands correspondence, p-adic group representations, local class field theory, Bruhat–Tits buildings* (Amiens, France)
 - 2023 **Teaching Training**, *ED MIIS (Caen, France)*

Skills

- Languages **French** (native), **English** (proficient user), **Brazilian Portuguese** (independent user), **Italian** (basic user)
- Programming **LaTeX**, **PARI/GP**, **Python**, **SageMath**, **Maxima**, **HTML**, **CSS**